

Yuting Zhou

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EDUCATION

Rutgers University , New Jersey, United States	Sep 2024 - Expected Jun 2026
• Major: M.S. Candidate in Computer Science	
Shanghai Institute of Technology , Shanghai, China	Sep 2020 - Jun 2024
• Major: Bachelor of Engineering in Software Engineering (GPA: 3.7, Top 10%)	

AWARDS & CERTIFICATES

- School-level Scholarship: 2nd Prize (2023, 2022), 3rd Prize (2022, 2021, 2020)
- Alibaba Cloud Certified Associate - Big Data, Huawei Certified ICT Associate - Intelligent Computing

RESEARCH EXPERIENCE

Research Assistant <i>Hierarchical Text Classification, Data Engineering, Language Models Training</i>	Jun 2025 - Present
RUTGERS Center for Advanced Infrastructure and Transportation (CAIT)	New Jersey, United States
• Processed 100k+ rail network records with automated Pandas scripts, improving anomaly detection efficiency.	
• Built Qt visualization tools that converts raw train data into intuitive graphs for efficient daily and location-based analysis.	
• Trained and fine-tuned hierarchical multi-label classifiers based on pretrained language models for incident categorization.	

INTERNSHIP EXPERIENCE

Front-end Engineer <i>JavaScript, Vue, React, WebSocket, MongoDB</i>	Jun 2023 - Aug 2023
Shanghai TENCHI Digital Tech Co., Ltd. (Tencent and Shanghai Metro Joint Venture)	Shanghai, China
• Contributed to reducing Shanghai Transit App page load latency by 30%, which improved retention.	
• Assisted in building a real-time lottery system supporting large-scale concurrent users with low latency via WebSocket and MongoDB.	
• Delivered 3 cross-team prototype interfaces by collaborating with PMs and iterating with Axure and Vue.	
Software Test Engineer <i>Pytest, JUnit, Selenium</i>	Jun 2022 - Sep 2022
Shanghai Spacecraft Electromechanical Equipment Co., Ltd.	Shanghai, China
• Built automated test environment across hardware configs with multi cases using Pytest/JUnit and CI pipeline.	
• Detected high-priority defects via Selenium UI automation, reducing production risk.	

PROJECT EXPERIENCE

Railway Incident Classification System <i>Hierarchical Learning, Data Preprocessing, Supervised Fine-tuning</i>	Aug 2025 - Present
• Built a DeBERTa-v3 hierarchical classifier for multi-level incident categorization on real-world logs.	
• Implemented end-to-end training and inference pipelines warm-up and encoder freezing.	
• Optimized training on CUDA / MPS / CPU with mixed precision and gradient accumulation.	
• Developed scalable data processing tools for multi-source merging, deduplication, and class consolidation.	
• Assessed performance and structured prediction using hierarchical exact match and per-level F1.	
Real-time Pose Estimation <i>OpenCV, PyTorch, TensorFlow</i>  github.com/Magicherry/Pose_Estimation	Sep 2023 - May 2024
• Achieved real-time skeletal detection at 30+ FPS with >80% exercise recognition accuracy via Google MoveNet.	
• Fine-tuned TensorFlow models to support 2 exercise types with avg. precision of 0.87.	
• Beta-tested with 50+ participants, gaining 70% positive engagement on the demo system.	
AyeNote Cloud Notes Platform <i>Vue 3, Django, MySQL, BeautifulSoup</i>  github.com/Magicherry/AyeNote	May 2022 - Apr 2023
• Improved web crawler parsing accuracy from 56% → 78% on 30k+ pages using async + SimHash.	
• Reduced per-note curation time by 60% by integrating LLM summarization.	
• Supported real-time collaboration for 20+ users with <250ms latency via Django Channels.	
King of Bots <i>Vue 3, Spring Boot, MySQL, WebSocket, Redis</i>  github.com/Magicherry/King-of-Bots	May 2022 - Nov 2022
• Developed a full-stack battle platform with Vue 3 + Spring Boot, supporting 50+ concurrent matches <250ms latency.	
• Designed 10+ secure REST APIs with JWT authentication and role-based access control.	

TECH STACK & LANGUAGES

Programming	Python, C++ , Java, Go
ML & NLP	PyTorch, Transformers, scikit-learn, TensorFlow, DeBERTa / BERT, LLM Fine-tuning, Hierarchical Classification
ML Systems	Mixed Precision Training, Gradient Accumulation, Multi-GPU / MPS Training, Experiment Tracking
Data & Analysis	NumPy, Pandas, Matplotlib, Seaborn, Feature Engineering
Databases	SQLite, MySQL, Redis, MongoDB
Backend	Django, FastAPI, Spring Boot, RESTful APIs, Microservices
Frontend	HTML, CSS, JavaScript, Vue 3, React, Next.js
DevOps	Linux, Docker, Kubernetes, GitHub Actions
Languages	English, Mandarin Chinese (Bilingual)